

Model fitting and detection power for the per-individual mixture-effect SNP model

W. R. P. Denault (derivation note for CHIMERA / SuSiE-Mix)

June 11, 2026

Conventions. One causal SNP, genotype $g_i \in \{0, 1, 2\}$ with Hardy–Weinberg frequencies $p_0 = (1 - f)^2$, $p_1 = 2f(1 - f)$, $p_2 = f^2$. Codings $x^A(g) = g$, $x^D(g) = \mathbf{1}\{g \geq 1\}$, $x^R(g) = \mathbf{1}\{g = 2\}$. Errors $\mathcal{N}(0, \sigma^2)$. Standardized effect $\theta = \beta/\sigma$; for an eQTL of variance-explained h^2 , $\theta = \sqrt{h^2/(2f(1 - f))}$, typically 0.05–0.3. All leading constants below are verified against numerical Kullback–Leibler (KL) minimization to ≥ 3 significant figures.

1 The model and its reduction to three genotype strata

Latent responder type $z_i \in \{A, D, R\}$, $P(z_i = m) = \pi_m$, independent of g_i , with

$$y_i \mid z_i=m, g_i \sim \mathcal{N}(x^m(g_i)\beta_m, \sigma^2), \quad p(y \mid g) = \sum_m \pi_m \phi_\sigma(y - x^m(g)\beta_m), \quad (1)$$

ϕ_σ the $\mathcal{N}(0, \sigma^2)$ density. Because g takes three values, the likelihood factorizes over the three conditional laws P_g , and these are the *entire* observable content:

$$\begin{aligned} P_0 &= \mathcal{N}(0, \sigma^2) & (\text{all codings vanish at } g=0: \text{no information}), \\ P_1 &= \pi_A \mathcal{N}(\beta_A, \sigma^2) + \pi_D \mathcal{N}(\beta_D, \sigma^2) + \pi_R \mathcal{N}(0, \sigma^2), \\ P_2 &= \pi_A \mathcal{N}(2\beta_A, \sigma^2) + \pi_D \mathcal{N}(\beta_D, \sigma^2) + \pi_R \mathcal{N}(\beta_R, \sigma^2). \end{aligned}$$

Each P_g is a Gaussian *location* mixture with common variance: the convolution of $\mathcal{N}(0, \sigma^2)$ with a discrete *mixing measure* $Q_g = \sum_m \pi_m \delta_{a_{gm}}$, $a_{gm} = x^m(g)\beta_m$. Write the central moments (cumulants) of Q_g as $c_{k,g}$, so P_g has mean $\bar{a}_g = c_{1,g}$, variance $\sigma^2 + c_{2,g}$, and cumulants $\kappa_k(P_g) = c_{k,g}$ for $k \geq 3$. Crucially, since every $a_{gm} = O(\theta\sigma)$,

$$c_{2,g} = O(\theta^2), \quad c_{3,g} = O(\theta^3), \quad c_{4,g} = O(\theta^4). \quad (2)$$

2 Fitting the three competing models

2.1 Additive (1 d.f.)

$y_i = \beta_0 + \alpha g_i + \varepsilon_i$. OLS gives $\hat{\alpha} = \widehat{\text{Cov}}(g, y) / \widehat{\text{Var}}(g)$, $\hat{\sigma}_{\text{add}}^2 = \frac{1}{n} \sum_i (y_i - \hat{\beta}_0 - \hat{\alpha} g_i)^2$. Bayesian single-effect Bayes factor (Wang et al. 2020), prior $\alpha \sim \mathcal{N}(0, \sigma_0^2)$, σ^2 plugged in: with $\hat{\alpha} = x^\top y / x^\top x$, $s^2 = \sigma^2 / x^\top x$ (x the centred additive coding),

$$\log \text{BF}_{\text{add}} = \frac{1}{2} \log \frac{s^2}{\sigma_0^2 + s^2} + \frac{1}{2} \hat{\alpha}^2 \left(\frac{1}{s^2} - \frac{1}{\sigma_0^2 + s^2} \right). \quad (3)$$

2.2 Genotypic / saturated mean (2 d.f.)

$\mathbb{E}[y \mid g] = \mu_g$ free. MLE: $\hat{\mu}_g = \bar{y}_g$ (stratum means); homoscedastic $\hat{\sigma}^2 = \frac{1}{n} \sum_g \sum_{i \in g} (y_i - \bar{y}_g)^2$; heteroscedastic $\hat{\sigma}_g^2 = \frac{1}{n_g} \sum_{i \in g} (y_i - \bar{y}_g)^2$. The genotypic mean space is spanned by $Z = [x^D \mid x^R]$ (dominant and recessive codings span the 2-d.f. deviation from the intercept). For $b \sim \mathcal{N}(0, V_0 I_2)$ and plugged-in σ^2 , the closed-form Bayes factor is

$$\log \text{BF}_{\text{geno}} = -\frac{1}{2} \log \det \left(I_2 + \frac{V_0}{\sigma^2} Z^\top Z \right) + \frac{1}{2\sigma^2} (Z^\top y)^\top \left(\frac{\sigma^2}{V_0} I_2 + Z^\top Z \right)^{-1} (Z^\top y). \quad (4)$$

(Equations (3)–(4) are the $q = 1, 2$ cases of the general Gaussian-prior linear-model BF; both match the exact multivariate-normal marginal likelihood to $\sim 10^{-13}$.)

2.3 Per-individual mixture (EM)

Treating z_i as missing, the complete-data log-likelihood is $\ell_c = \sum_i \sum_m \mathbf{1}\{z_i=m\} [\log \pi_m - \frac{1}{2\sigma^2} (y_i - x^m(g_i)\beta_m)^2] + \text{const.}$

E-step.

$$w_{im} = \frac{\pi_m \phi_\sigma(y_i - x^m(g_i)\beta_m)}{\sum_{m'} \pi_{m'} \phi_\sigma(y_i - x^{m'}(g_i)\beta_{m'})}. \quad (5)$$

M-step (maximize $\mathbb{E}[\ell_c]$ under $\sum_m \pi_m = 1$):

$$\pi_m = \frac{1}{n} \sum_i w_{im}, \quad \beta_m = \frac{\sum_i w_{im} x^m(g_i) y_i}{\sum_i w_{im} x^m(g_i)^2}, \quad \sigma^2 = \frac{1}{n} \sum_{i,m} w_{im} (y_i - x^m(g_i)\beta_m)^2. \quad (6)$$

The β_m updates are weighted least squares; the algorithm increases the observed log-likelihood $\ell = \sum_i \log [\sum_m \pi_m \phi_\sigma(y_i - x^m(g_i)\beta_m)]$ monotonically. Under the null $\pi_D = \pi_R = 0$ the parameters β_D, β_R are *unidentified*, which drives the non-regularity in §4.4.

3 What is identifiable, and at which order

Let $\mathcal{M}_{\text{add}} \subset \mathcal{M}_{\text{geno}}^{\text{hom}} \subset \mathcal{M}_{\text{geno}}^{\text{het}}$ denote the model classes. For a truth P^* generated by (1), the per-observation expected log-likelihood ratio between the best fit in the mixture model ($\text{KL} = 0$) and the best fit in a null class $\mathcal{M}_0$ is exactly the *separation information*

$$I_{\text{sep}}(\mathcal{M}_0) = \min_{Q \in \mathcal{M}_0} \text{KL}(P^* \parallel Q) = \sum_g p_g \min_{Q_g \in \mathcal{M}_0} \text{KL}(P_g \parallel Q_g). \quad (7)$$

Lemma 1 (Gaussian KL decomposition). *For any distribution P with mean μ , variance v , and a Gaussian $\mathcal{N}(m, s)$,*

$$\text{KL}(P \parallel \mathcal{N}(m, s)) = J(P) + \frac{1}{2} \left(\frac{v}{s} - 1 - \log \frac{v}{s} \right) + \frac{(\mu - m)^2}{2s},$$

where $J(P) = \text{KL}(P \parallel \mathcal{N}(\mu, v))$ is the negentropy, with the Edgeworth expansion $J(P) = \frac{\kappa_3(P)^2}{12 v^3} + \frac{\kappa_4(P)^2}{48 v^4} + O(\kappa^3)$.

The three terms isolate *shape*, *variance*, and *mean* mismatch. Minimizing over each null class and using (2) gives:

Proposition 1 (Detection-information hierarchy). *As $\theta \rightarrow 0$,*

$$I_{\text{sep}}(\mathcal{M}_{\text{add}}) = \frac{1}{2\sigma^2} \sum_g p_g r_g^2 + O(\theta^4) = \Theta(\theta^2), \quad (8)$$

$$I_{\text{sep}}(\mathcal{M}_{\text{geno}}^{\text{hom}}) = \frac{1}{4} \text{Var}_p\left(\frac{c_{2,g}}{\sigma^2}\right) + O(\theta^6) = \Theta(\theta^4), \quad (9)$$

$$I_{\text{sep}}(\mathcal{M}_{\text{geno}}^{\text{het}}) = \sum_g p_g \frac{c_{3,g}^2}{12\sigma^6} + O(\theta^8) = \Theta(\theta^6) \quad (\Theta(\theta^8) \text{ if } c_{3,g} \equiv 0), \quad (10)$$

where r_g is the weighted-least-squares residual of the stratum means $\{\bar{a}_g\}$ against a function linear in g (so $\sum_g p_g r_g^2$ is the dominance deviation $d = \bar{a}_2 - 2\bar{a}_1 + \bar{a}_0$, up to a positive constant), $c_{2,g} = \pi(1 - \pi)\Delta_g^2$ is the excess within-genotype variance with Δ_g the spread of component means, and Var_p is over genotype frequencies.

Proof sketch. By Lemma 1. (i) $\mathcal{M}_{\text{add}}$ constrains the means to be affine in g and shares a common variance; the leading term is the mean mismatch $\sum_g p_g (\bar{a}_g - m_g)^2 / (2\sigma^2)$, minimized by WLS, giving (8); the dominance deviation is $O(\theta)$ so this is $\Theta(\theta^2)$. (ii) $\mathcal{M}_{\text{geno}}^{\text{hom}}$ matches the means exactly, leaving the variance-mismatch term; with $v_g/\sigma^2 = 1 + \delta_g$, $\delta_g = c_{2,g}/\sigma^2 = O(\theta^2)$, expand $\frac{1}{2}(\frac{v_g}{s} - 1 - \log \frac{v_g}{s}) \approx \frac{1}{4}(\delta_g - \bar{\delta})^2$ and minimize the common variance s over $\bar{\delta} = \sum_g p_g \delta_g$, giving $\frac{1}{4} \text{Var}_p(\delta_g) = \Theta(\theta^4)$. (iii) $\mathcal{M}_{\text{geno}}^{\text{het}}$ matches mean and variance per stratum, leaving only the negentropy $J(P_g) = c_{3,g}^2/(12\sigma^6) + O(\theta^8)$, which is $\Theta(\theta^6)$ (and $\Theta(\theta^8)$ when the mixing is symmetric, e.g. $\pi = \frac{1}{2}$, where $c_{3,g} = 0$). $\square \square$

Numerical verification of the constants (not just the orders). Two-type truth (additive responders weight $1 - \pi$, effect θ ; recessive responders weight π , effect $c_R\theta$), $\pi = 0.3$, $c_R = 2$, $f = 0.4$:

θ	I_{add} num	pred	I_{hom} num	pred	I_{het} num	pred
0.05	2.593×10^{-5}	2.592×10^{-5}	1.719×10^{-8}	1.720×10^{-8}	4.40×10^{-12}	4.41×10^{-12}
0.10	1.038×10^{-4}	1.037×10^{-4}	2.749×10^{-7}	2.752×10^{-7}	2.81×10^{-10}	2.82×10^{-10}
0.20	4.173×10^{-4}	4.147×10^{-4}	4.384×10^{-6}	4.403×10^{-6}	1.77×10^{-8}	1.81×10^{-8}
0.30	9.460×10^{-4}	9.331×10^{-4}	2.208×10^{-5}	2.229×10^{-5}	1.97×10^{-7}	2.06×10^{-7}

4 Detection power and required sample size

4.1 Likelihood-ratio test against a fixed null

Let $\Lambda_n = \ell_n(\hat{\vartheta}_{\mathcal{M}_1}) - \ell_n(\hat{\vartheta}_{\mathcal{M}_0})$. When the truth lies in $\mathcal{M}_1$ (mixture), the law of large numbers gives $n^{-1}\Lambda_n \rightarrow I_{\text{sep}}(\mathcal{M}_0)$ a.s., hence

$$\mathbb{E}[2\Lambda_n] \approx 2n I_{\text{sep}}(\mathcal{M}_0). \quad (11)$$

Check: $\mathcal{M}_0 = \mathcal{M}_{\text{geno}}^{\text{hom}}$, $\theta = 0.3$: (11) predicts $2nI_{\text{hom}} = 2.23$ at $n = 5 \times 10^4$; direct simulation of the EM-LRT gives mean 2.29.

For a regular nested comparison with r extra degrees of freedom, under local alternatives $2\Lambda_n \sim \chi_r^2(\lambda)$ with noncentrality $\lambda = 2n I_{\text{sep}}$. To attain power $1 - \beta$ at level α one needs $\lambda \geq \lambda_{\alpha,\beta,r}^*$, hence

$$\boxed{n^* = \frac{\lambda_{\alpha,\beta,r}^*}{2 I_{\text{sep}}(\mathcal{M}_0)}}, \quad \lambda_{\alpha,\beta,1}^* \approx (z_{1-\alpha} + z_{1-\beta})^2. \quad (12)$$

Genome-wide screen ($\alpha = 5 \times 10^{-8}$, $\beta = 0.2$, $r = 1$): $\lambda^* \approx (5.45 + 0.84)^2 \approx 39.6$, so $n^* \approx 19.8/I_{\text{sep}}$.

4.2 The decisive distinction: “non-additive” vs. “a mixture”

Equation (8) shows non-additivity is detected at $\Theta(\theta^2)$: the dominance deviation. **This does not establish a per-individual mixture**—a single deterministic dominance/recessive dose-response produces the same mean structure. The mixture-specific content is the *within-genotype overdispersion / non-Gaussianity*, detectable only against $\mathcal{M}_{\text{geno}}$, i.e. at $\Theta(\theta^4)$ (homoscedastic null) or $\Theta(\theta^6)$ (heteroscedastic null, the honest confounder). Substituting the verified I_{sep} into (12):

θ	n^* vs additive	n^* vs genotypic (hom.)	n^* vs genotypic (het.)
0.10	$\sim 1.9 \times 10^5$	$\sim 7.2 \times 10^7$	$\sim 7 \times 10^{10}$
0.20	$\sim 4.8 \times 10^4$	$\sim 4.5 \times 10^6$	$\sim 1.1 \times 10^9$
0.30	$\sim 2.1 \times 10^4$	$\sim 8.9 \times 10^5$	$\sim 9.6 \times 10^7$

For GTEx whole blood ($n \approx 670$) the mixture-vs-genotypic test is short of the required n by 3–5 (hom.) or 5–8 (het.) orders of magnitude. A direct EM-LRT simulation confirms 0% power at $n = 670$, rising to only $\sim 25\%$ at $n = 5 \times 10^4$ for $\theta = 0.3$; and a non-mixture heteroscedastic-genotypic locus triggers the same statistic, producing false “mixture” calls.

4.3 Locally most powerful test for the $\Theta(\theta^4)$ signal

Near the additive boundary the leading mixture departure from $\mathcal{M}_{\text{geno}}^{\text{hom}}$ is the heterogeneity of within-genotype variances $\delta_g = c_{2,g}/\sigma^2$. A Levene/Bartlett-type score test on the stratum variances has noncentrality, to leading order,

$$\lambda = 2n I_{\text{sep}}(\mathcal{M}_{\text{geno}}^{\text{hom}}) = \frac{n}{2} \text{Var}_p\left(\frac{c_{2,g}}{\sigma^2}\right), \quad \text{power} \approx \Phi\left(\sqrt{\lambda} - z_{1-\alpha}\right), \quad (13)$$

giving $n^* = \Theta(\theta^{-4})$, consistent with (9). This test is regular (the variance components are identified) and avoids fitting the unidentified mixture; it should be the workhorse, calibrated by simulation.

4.4 Non-regularity of the mixture LRT

Testing $H_0 : \pi_D = \pi_R = 0$ vs. the mixture is a Hartigan/Davies problem: under H_0 the effects β_D, β_R are unidentified and the mixing weights sit on the simplex boundary. Consequently $2\Lambda_n$ does *not* converge to a χ^2 ; it behaves like the supremum of a Gaussian process and can drift like $\log \log n$ (Hartigan 1985; Chen & Chen 2001; Liu & Shao 2003). Two consequences: (a) analytic χ^2/BIC calibration of a mixture test is invalid, so thresholds must be obtained by simulation under $\mathcal{M}_{\text{geno}}$ (including a heteroscedastic null); (b) any Bayes factor is strongly sensitive to the prior on (β_D, β_R) —improper priors yield arbitrary-scale, meaningless $\text{BF}_{\text{mix:geno}}$. Hence informative, scale-matched effect priors and a Dirichlet/spike-and-slab prior on π are required, and the data override the additive default only once $n\theta^4 \gtrsim \lambda_{\text{prior}}$.

5 Conclusion

Coding selection (additive/dominant/recessive; heterogeneous regulation) is a *mean-structure* problem, identified at $\Theta(\theta^2)$ —this is the well-posed core that SuSiE-Mix exploits. The *per-individual mixture interpretation* is a higher-moment problem, identified only at $\Theta(\theta^4)$ – $\Theta(\theta^6)$ relative to a genotypic model, hence near-non-identifiable at single loci for GTEx-scale data and confounded with ordinary heteroscedasticity. The correct discriminating statistic is $\text{BF}_{\text{mix:geno}}$ (or the variance-heterogeneity score test), simulation-calibrated against a heteroscedastic genotypic null, with the

explicit sample-size requirement (12). Lead empirical claims with testable only in large biobank traits or via cross-locus partial pooling.